

Climatrix AI Geospatial Intelligence for Flood Risk

Giving people time before disaster strikes in Lagos, Nigeria.



The Challenge: Lagos Flooding

Lagos faces increasing annual flooding, causing significant economic and social disruption. In 2024, **9,324 people** were affected in Lagos Mainland, with **409 houses** damaged.

Minimize Project Delays

Improve site selection for the construction industry.

Reduce Investment Risks

Increase buyer transparency for the real estate sector.

Enable Proactive Planning

Efficiently allocate emergency resources for government agencies.

Data & Methodology

We combine historical and real-time data from diverse sources to provide a comprehensive flood risk picture.

Key Data Sources:

- **Real Estate Data:** PropertyPro listings.
- **Weather & Climate Data:** Open-Meteo API (current, 7-day forecast, historical(2020-2024)).
- **Historical Flood Data:** NEMA (2013-2022).
- **Geospatial Data:** OpenStreetMap (water proximity), Open-Elevation API (elevation), QGIS (drainage density, LGA boundaries).
- **Demographic Data:** WorldPop.org (2020 population).

Data Utilization:

Datasets are cleaned and joined by LGA for cross-table analysis, enabling calculation of key metrics:

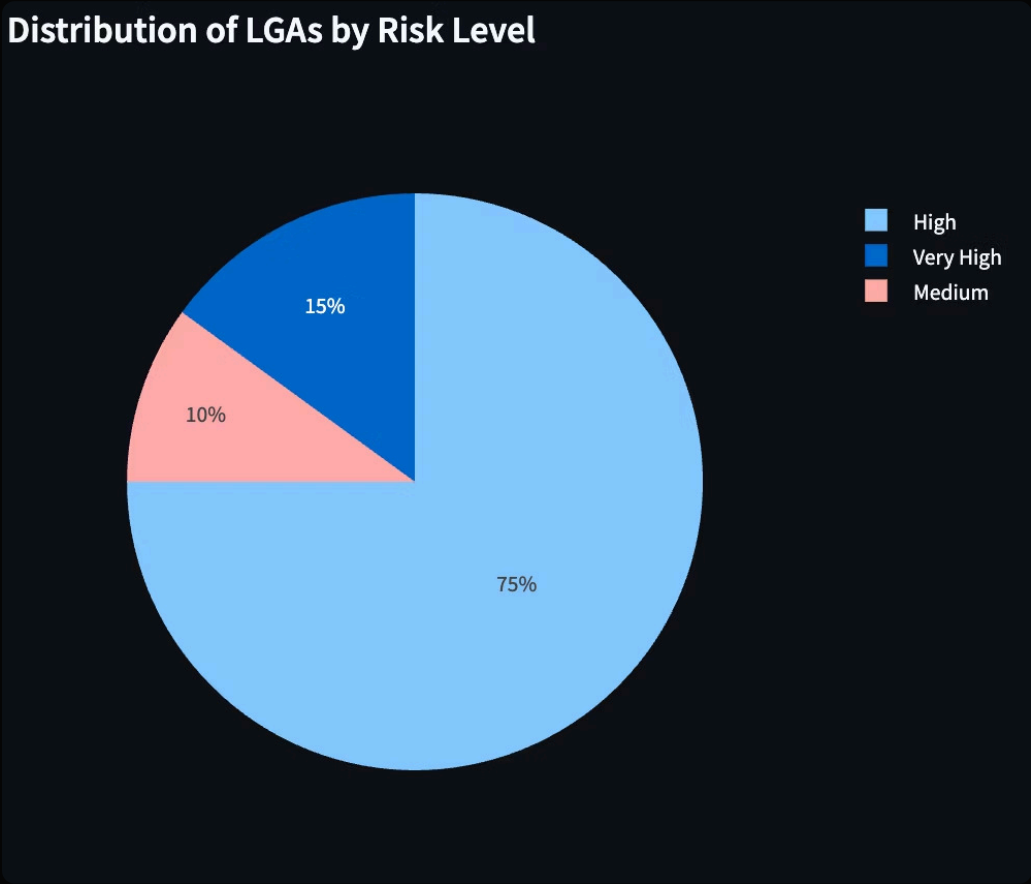
- **Advanced Risk Score:** Likelihood and impact of flooding.
- **Vulnerability Index:** Potential human impact (risk + population).
- **Investment Opportunity Index:** Identifies undervalued real estate markets.
- **Drainage Inefficiency Score:** Assesses infrastructure's ability to handle rainfall.

Exploratory Analysis: Lagos Flood Overview

The average flood risk score for Lagos LGAs is **7.15**, with **90%** falling into "High" or "Very High" risk categories.



Ibeju Lekki, Eti-Osa, and Lagos Island have the highest risk scores. **Alimosho, Oshodi/Isolo, and Kosofe** are the most populated areas.



75% of LGAs are "High" risk, **15%** are "Very High" risk, and only **10%** are "Medium" risk, highlighting pervasive vulnerability.



Sector Solution

Our solution provides proactive intelligence, enabling strategic preparedness across sectors.

Solution: Government & Urban Planning

Climatrix AI provides tools for strategic planning, resource allocation, and citizen protection.



Interactive Risk Map

Color-coded map by Advanced Risk Score for immediate visual understanding of risk concentration.



Population Vulnerability Analysis

Maps Vulnerability Index (risk + population) to prioritize resources where human impact is greatest.



LGA Infrastructure Analysis

Interactive dashboard displaying Flood Risk, Population Vulnerability, and Drainage Inefficiency for precise local decision-making.



Drainage Inefficiency Hotspot Ranking

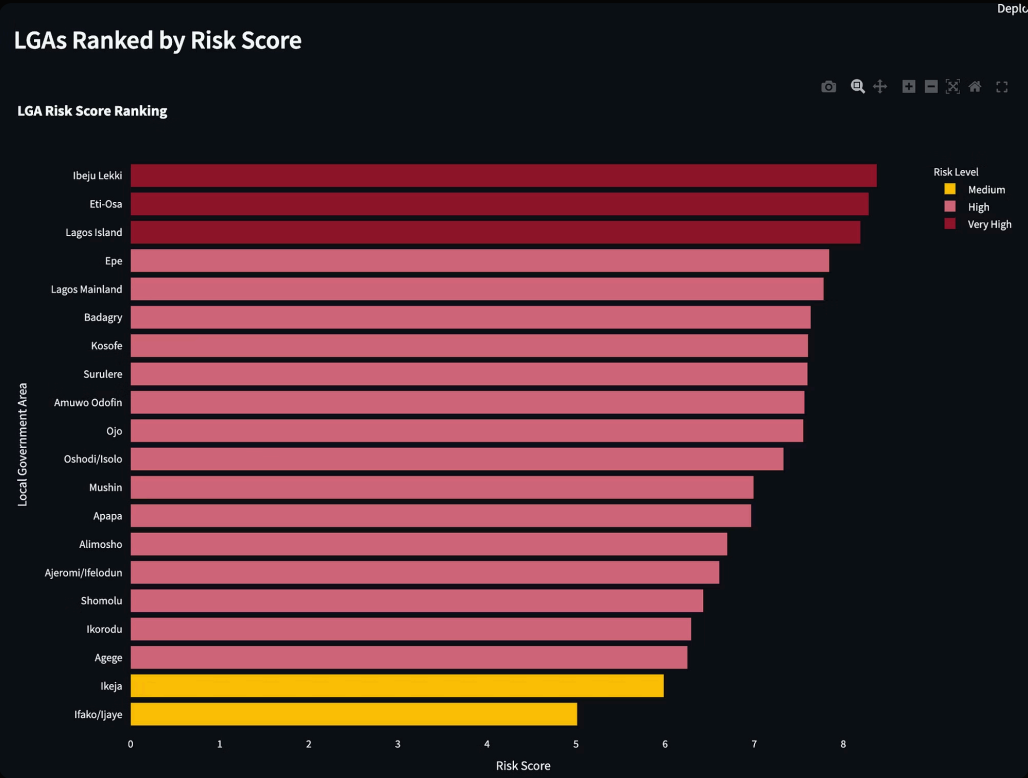
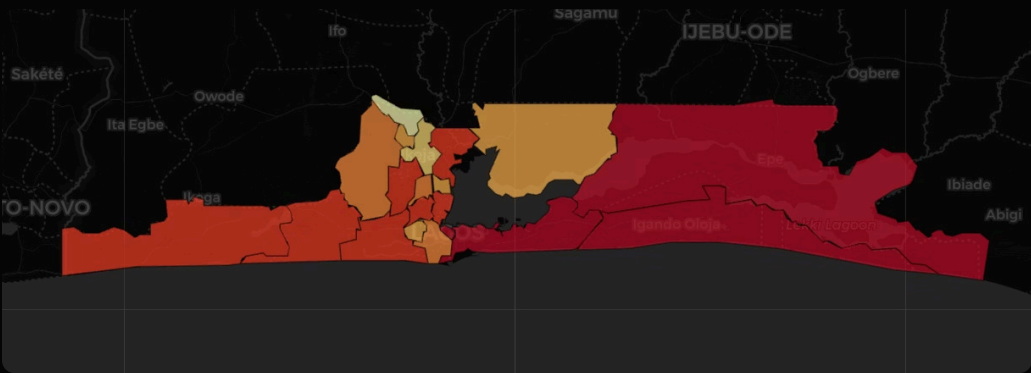
Ranks LGAs by drainage inefficiency, providing a clear priority list for infrastructure intervention.

Interactive Risk Map: Visualizing Flood Risk

Climatrix AI's Interactive Risk Map dynamically visualizes flood risk across Lagos.

The map color-codes LGAs by their Advanced Risk Score. Users can hover for specific scores and population density, enabling immediate visual understanding for proactive decision-making.

The Lagos State Ministry of the Environment can use this map to identify critical LGAs for immediate investment in flood defense, optimizing resource allocation.

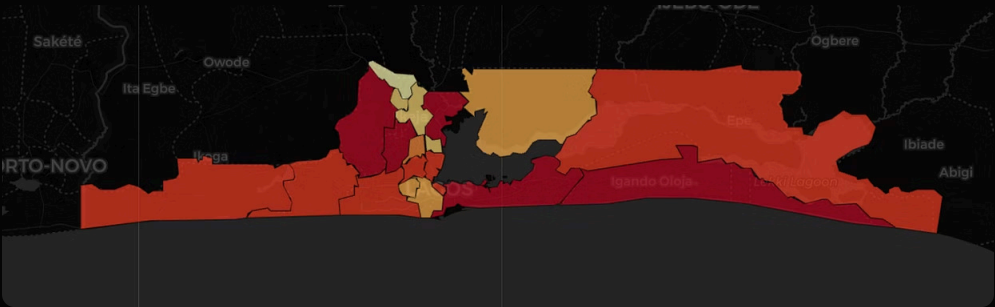


Population Vulnerability Analysis

This feature uses the **Vulnerability Index**, which combines flood risk and population density. It helps identify communities most at risk.

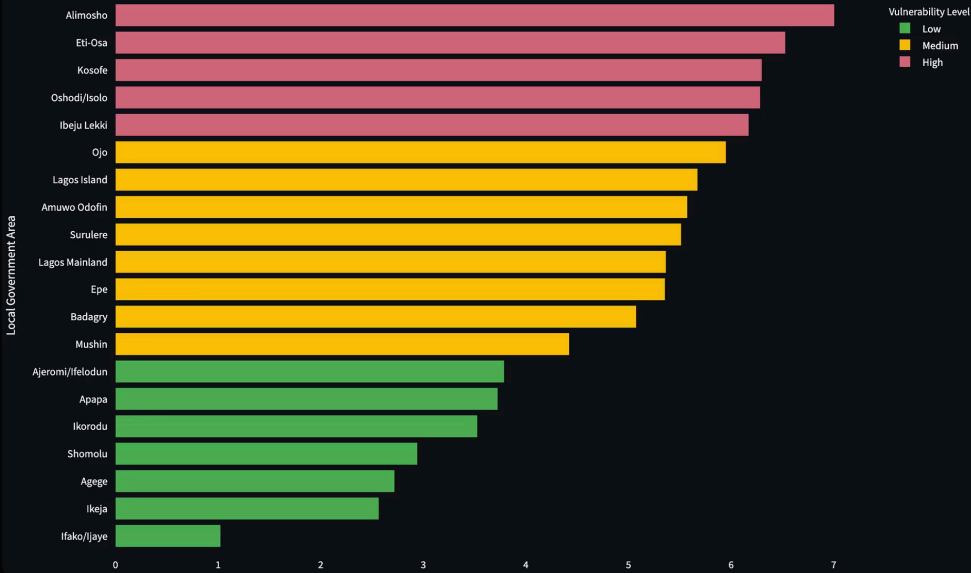
This directs resources to high-risk areas with many people, ensuring fair protection.

The Lagos State Emergency Management Agency (LASEMA) can use this analysis to set up emergency shelters and plan evacuation routes in high-vulnerability areas before the rainy season.



This chart ranks LGAs by their vulnerability score, ideal for prioritizing mitigation projects and emergency shelter placement.

LGA Vulnerability Ranking



Interactive LGA Infrastructure Analysis

This dashboard offers a comprehensive, color-coded overview of each LGA's **Flood Risk**, **Population Vulnerability**, and **Drainage Inefficiency Scores**. It empowers agencies like the Public Works Corporation to make evidence-based decisions, optimizing crucial infrastructure investments such as culvert upgrades.

Government & Urban Planning

LGA Infrastructure & Vulnerability Analysis

Select an LGA to view its key risk, vulnerability, and drainage metrics. This is crucial for prioritizing infrastructure investment and resource allocation.

Select an LGA for detailed analysis:

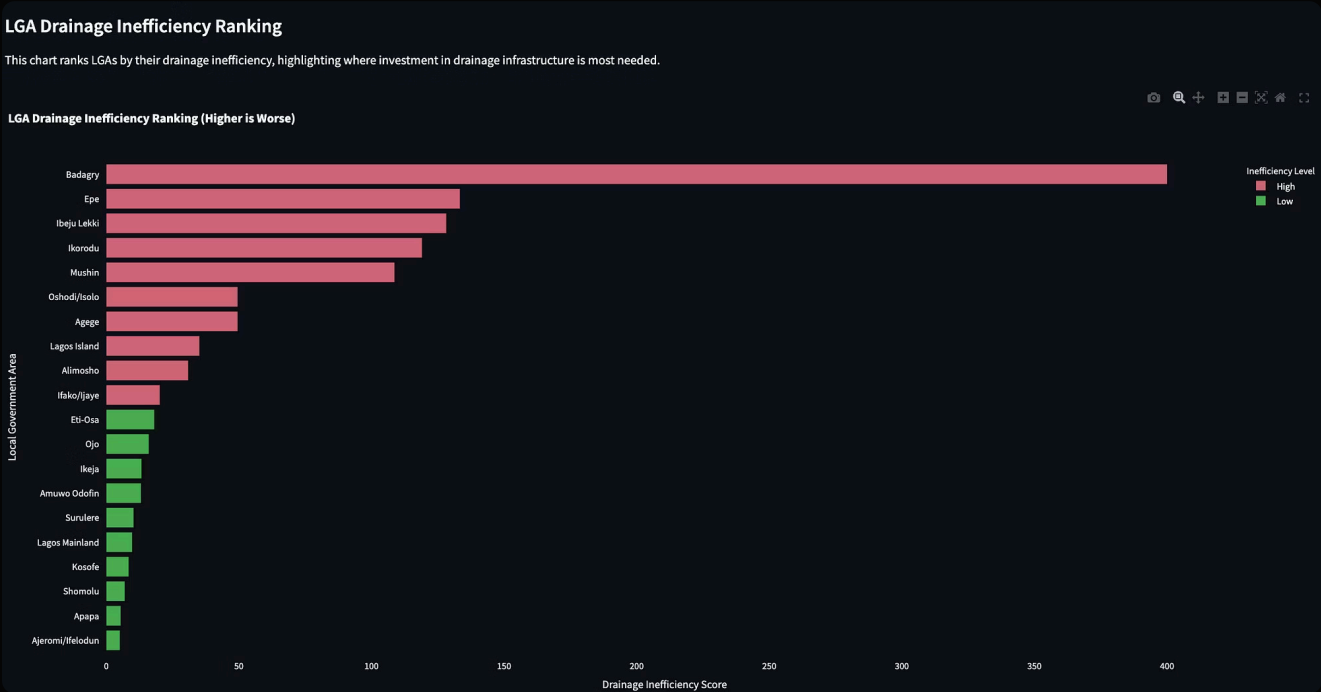
Ikeja

Metrics for Ikeja

Flood Risk Score 5.98 Status: Medium	Vulnerability Score 2.56 Status: Low	Drainage Inefficiency 13.26 Level: Low
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Drainage Inefficiency Hotspot Ranking

A dynamic bar chart ranks LGAs by their **Drainage Inefficiency Score**, highlighting areas most prone to water accumulation. This data-driven list prioritizes infrastructure intervention, enabling the Ministry of Environment to pinpoint high-priority LGAs for drain improvement programs.



Solution: Real Estate Sector

Leverage Climatrix AI for smarter, safer, and more profitable investment decisions.

1

Interactive Investment Opportunity Analysis

Identifies undervalued areas where property prices may not fully reflect lower flood risk.

2

Risk-Adjusted Investment Ranking

Ranks LGAs by Investment Opportunity Index, providing a strategic overview of the Lagos market.

3

Lowest Risk Property Marketing

Queries properties in low/medium risk LGAs, highlighting safety as a key selling point for buyers.

Interactive Investment Opportunity Analysis

This interactive tool helps investors identify "High Potential" areas by instantly viewing an LGA's **Average Property Value**, **Flood Risk Score**, and its unique **Investment Opportunity Index** (Price / Risk Score).

It's a powerful due diligence tool to uncover undervalued, lower-risk investments.

Investment Opportunity Analysis

Select an LGA to view its investment profile. This analysis helps identify potentially undervalued areas where property prices may not fully reflect their lower flood risk.

Select an LGA to analyze:

Agege

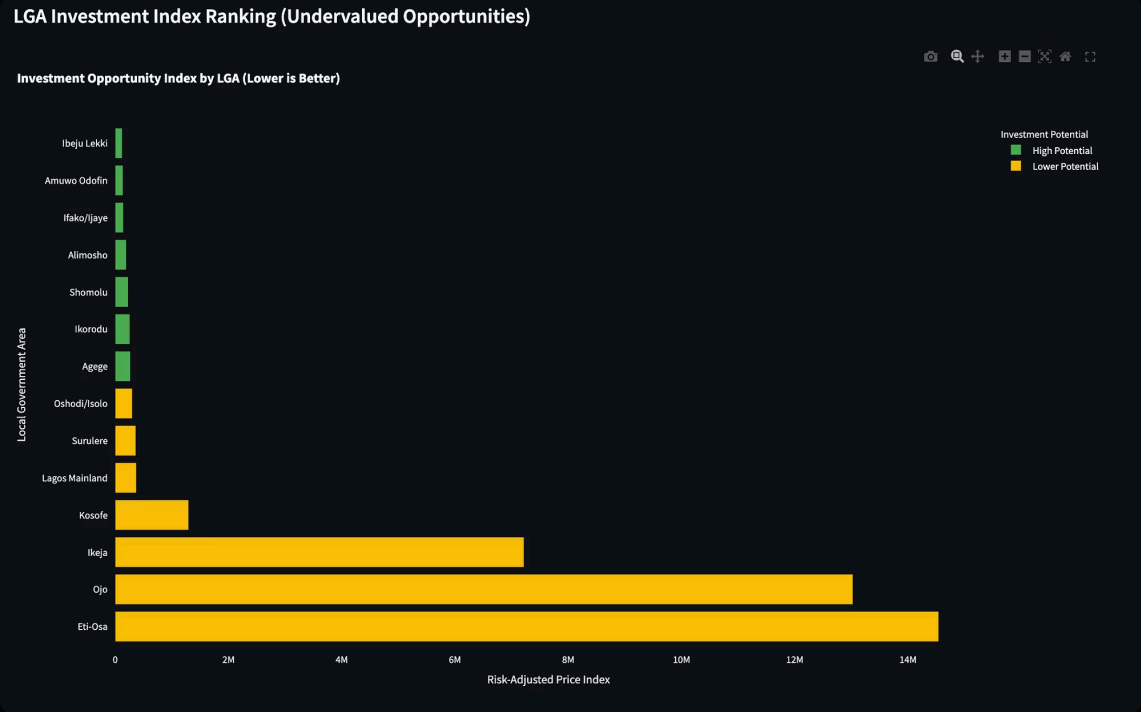
Investment Profile for Agege

Average Property Value (M)	Investment Opportunity Index ⓘ	Flood Risk Score (0-10)
1,650,000	264,019.13	6.25
	Potential: High Potential	Status: High

Risk-Adjusted Investment Ranking

This bar chart ranks LGAs by Investment Opportunity Index, highlighting "High Potential" areas in green. It offers developers and investors a strategic overview of the Lagos market for data-driven, risk-adjusted acquisition strategies.

For instance, a fund manager can quickly identify the top 5 LGAs for financially sound investment opportunities, guiding their quarterly investment strategy.



Lowest Risk Property Marketing

This feature queries properties in "Low" or "Medium" risk LGAs, highlighting safety as a key selling point. It provides a direct marketing advantage for real estate agents, building trust and potentially commanding higher prices.

For example, a marketing firm can use this list for an "Invest with Peace of Mind" campaign, attracting risk-averse buyers with verifiable "Risk Scores."

Top 5 Properties by Lowest Risk (Marketing Advantage)

The following properties are located in the LGAs with the lowest calculated flood risk, making them ideal for marketing as safe investments.

A Room Self Contained in Ifako/Ijaye | Price: ₦400,000 | Risk Score: 5.01 (Medium)

Newlybuilt Miniflat Apartment, At Meiran Abule Egba in Ifako/Ijaye | Price: ₦1,000,000 | Risk Score: 5.01 (Medium)

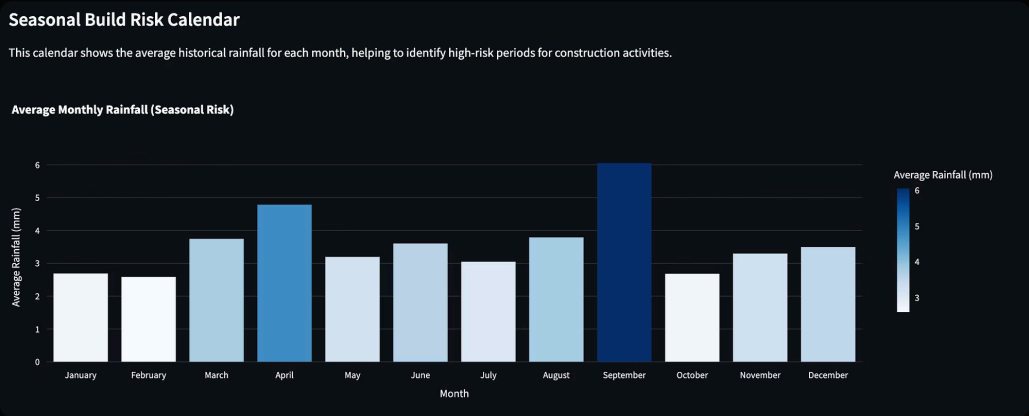
2 Bedroom Apartment in Ikeja | Price: ₦14,000,000 | Risk Score: 5.98 (Medium)

3 Bedroom Apartment in Ikeja | Price: ₦3,000,000 | Risk Score: 5.98 (Medium)

3 Bedroom Flat in Ikeja | Price: ₦6,000,000 | Risk Score: 5.98 (Medium)

Solution: Construction Sector

Mitigate weather-related risks, optimize schedules, and build more resilient structures.



Seasonal Build Risk Calendar

Shows average monthly rainfall, helping schedule weather-sensitive work e.g., Ikeja projects finish foundations before April to avoid June–Sept peak rains.

Resilient Design Recommendations

Select an LGA to receive an auto-generated checklist of design recommendations based on its specific flood risk level.

Select an LGA for Design Recommendations

Lagos Mainland

Risk Level for Lagos Mainland: High

- ✓ **Foundation:** Minimum 2.9m depth required.
- ✓ **Drainage:** Enhanced drainage systems (e.g., French drains) recommended.
- ✓ **Materials:** Use of flood-resistant materials is strongly advised.
- ✓ **Electrical:** Elevate electrical systems at least 1.5m above ground level.

Resilient Design Recommendations

Generates a checklist of building design recommendations based on LGA-specific risk levels, standardizing resilient construction practices.

Cross-Sector Solution: Predictive Early Warning System

Transforms users from reactive to prepared, providing proactive, forward-looking intelligence for all sectors..

Predictive Analytics & Early Warning for Critical Zones

This page focuses exclusively on LGAs identified as having a 'Very High' risk level to provide targeted, actionable alerts.

Identified Critical Zones: Lagos Island, Ibeju Lekki, Eti-Osa

7-Day Flood Early Warning System

High Risk ALERT: Significant rainfall predicted in critical zones within the next 7 days.

High Risk for Eti-Osa

- Wednesday, August 06: Predicted rainfall of 3.40 mm
- Friday, August 08: Predicted rainfall of 10.60 mm
- Saturday, August 09: Predicted rainfall of 4.20 mm

High Risk for Ibeju Lekki

High Risk for Lagos Island

7-Day Precipitation Forecast for a Critical Zone

Select a Critical LGA for 7-Day Forecast:

Lagos Island

7-Day Precipitation Forecast for Lagos Island

Date	Predicted Rainfall (mm)
Aug 6 2025	3.40
Aug 7	2.20
Aug 8	10.60
Aug 9	4.20
Aug 10	1.50
Aug 11	1.50
Aug 12	0.80

Made with GAMMA

Integrated Use Case Scenarios

Government

LASEMA receives 4-day advance alert for Mushin, pre-deploys rescue teams, and verifies drainage.



Real Estate

Facility manager in Apapa gets 7-day forecast, schedules inspections, tests generators, and notifies tenants.



Construction

Site manager in Ikeja sees 2-day rain prediction, postpones concrete pour, secures materials, and arranges de-watering pumps.

Recommendations for Future Development

1 Develop Interactive Thematic Maps

Create choropleth maps for Investment Opportunity and Drainage Inefficiency to visualize geographic clusters and hotspots.

2 Create Sector-Specific Dashboards

Build dedicated, subscription-based portals with specialized metrics for each sector.

3 Integrate Real-Time Alerts

Implement SMS & Email notifications for subscribers in "Very High" risk zones during predicted heavy rainfall.

4 Expand Geographically

Scale the platform to other flood-prone Nigerian states to capture a larger national market.